Design of Enhanced AMHS with Color Sorting Integration Using 3D Printing Technology

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Abstract:

Material handling is a significant challenge in industries, often requiring substantial manpower and labor. To address this, we propose a color sorting mechanism that leverages advanced sensors and robotics to classify and separate materials based on color. This technology is extensively used in industries such as recycling, agriculture, food processing, and manufacturing, where precise sorting enhances quality control and reduces waste. By minimizing human intervention, the system improves accuracy, reduces errors, and optimizes production efficiency. Additionally, the integration of conveyor-based or robotic sorting mechanisms enhances automation, making industrial operations more sustainable and cost-effective.

This prototype is designed using Fusion 360 and 3D printing for structural components, while the electronics are controlled by an Arduino with three degrees of freedom achieved through servo motors, an electromagnetic end effector. Incorporating Smart Warehousing and Industry 4.0 principles, the system enhances automation in material handling. The color-sorting mechanism utilizes advanced sensors to detect subtle color variations, enabling high-speed and precise material classification. This approach optimizes efficiency, reduces human intervention, and improves accuracy in industrial applications such as recycling, manufacturing, and logistics.

Keywords: SolidWorks, TCS 34725 Sensor, Articulated robot, Servo motor

Introduction:

The field of robotics is a branch of engineering and computer science that deals with the design, construction, and operation of robots. It is an interdisciplinary field that draws upon knowledge and techniques from various other fields, including mechanical engineering, electrical engineering, computer science, and artificial intelligence. The goal of robotics is to create machines that can perform tasks autonomously, with minimal human intervention. This includes tasks that are too dangerous, too difficult, or too repetitive for humans to perform. Robotics has already had a significant impact in areas such as manufacturing, space exploration, and healthcare, and it is expected to have an even greater impact in the future. One of the key challenges in robotics is developing machines that can interact with their environment and adapt to changing conditions. This requires the use of sensors and other technologies that allow robots to perceive and interpret their surroundings. Robotics also involves the development of algorithms and software that allow robots to make decisions and carry out tasks autonomously.

Conventional grippers struggle to efficiently sort metallic waste due to variations in shape, size, and weight, leading to slower processing, increased error rates, and potential damage to robotic arms and surrounding equipment. Their limited flexibility restricts handling diverse objects, reducing throughput and overall efficiency, while inadequate gripping poses safety risks by causing dropped objects that endanger personnel and machinery. Additionally, the TCS3200 color sensor has a narrow detection range and is sensitive to ambient light, making it prone to misclassification errors in metallic waste sorting. Unlike advanced sensors like the TCS34725, it struggles with subtle color variations and fluctuating lighting conditions, leading to inconsistent readings that compromise sorting accuracy and efficiency. These limitations collectively reduce the reliability of automated sorting systems and impact the quality of material segregation.